

Lecture 7: Generalization I

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Generalization in machine learning refers to the ability of a model to make accurate predictions on new, unseen data. Generalization is crucial for machine learning models because in real-world applications, the model will be exposed to new data that it has not seen before. If a model does not generalize well, it will not be able to make accurate predictions in these situations, and its performance will be limited. In this lecture, we will discuss generalization bounds which are important in deep learning because they provide a theoretical framework for understanding and quantifying the expected error of a model on new, unseen data. These bounds give us insight into how well a model will perform in practice.

Let's introduce the problem formally. We consider a hypothesis space as a set $\mathcal{F}$ of functions from $\mathcal{X} \subseteq \mathbb{R}^d$ to $\mathcal{Y} \subseteq \mathbb{R}$. Given a data distribution $\mathcal{D}$ over $(\mathcal{X}, \mathcal{Y})$, we define the true risk of a function $f \in \mathcal{F}$ as

$$R(f) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}}[\ell(f(\mathbf{x}), y)],$$

where $\ell(\cdot, \cdot)$ is a given loss function. We will mostly discuss a classification setting where $\ell(f(\mathbf{x}), y) = \mathbf{1}_{f(\mathbf{x}) \neq y}$.

Since we typically do not have access to the exact data distribution but only a sample set $S = \{(\mathbf{x}_i, y_i)\}_{i=1}^n \sim D$, we also define the empirical risk as

$$R_S(f) = \frac{1}{n} \sum_{(x, y) \in S} [\ell(f(\mathbf{x}), y)].$$

The typical form of a generalization bound we are interested in is:

$$\underbrace{R(f)}_{\text{True risk}} \leq \underbrace{R_S(f)}_{\text{Empirical risk}} + \underbrace{m(\text{complexity of class of functions}, n)}_{\text{(a function that approaches 0 as } n \text{ approaches infinity)}},$$

where n is the number of training datapoints and m is a function that measures the complexity of a class of functions, such as the VC dimension, or the Rademacher complexity, which we will discuss in this lecture.

1 Basic Concentration Bound

We can obtain a basic bound by applying Hoeffding's inequality to a given function class and then using the union bound over the set of function classes.

In the following, we fix the function $f \in \mathcal{F}$ and let $Z_i = (x_i, y_i)$ be i.i.d. random variables such that $\ell(Z) \in [a, b]$. The quantity we are interested in is

$$R(f) - R_S(f) = \mathbb{E}[\ell(Z)] - \frac{1}{n} \sum_{i=1}^n \ell(Z_i).$$

Using Hoeffding's inequality, as well as the independence of the random variables $\ell(Z_i)$ ¹ we get:

$$P \left[\left| \mathbb{E}[\ell(Z)] - \frac{1}{n} \sum_{i=1}^n \ell(Z_i) \right| > \epsilon \right] \leq 2 \exp \left(-\frac{2n\epsilon^2}{(b-a)^2} \right).$$

¹Recall that if $\{Z_i\}_i$ are i.i.d. variables, then $\{\ell(Z_i)\}_i$ are also i.i.d. as long as ℓ is a Borel-measurable function, which includes all functions of practical interest in machine learning.

Setting the right-hand side to be δ , we obtain:

$$\begin{aligned}
 \delta &= 2 \exp \left(-\frac{2n\epsilon^2}{(b-a)^2} \right) \\
 \implies \log \delta &= \log 2 - \left(\frac{2n\epsilon^2}{(b-a)^2} \right) \\
 \implies \log \frac{1}{\delta} &= -\log 2 + \left(\frac{2n\epsilon^2}{(b-a)^2} \right) \\
 \implies \epsilon^2 &= \frac{(b-a)^2}{2n} \left(\log 2 + \log \frac{1}{\delta} \right) \\
 \implies \epsilon &= (b-a) \sqrt{\frac{\log \frac{2}{\delta}}{2n}}.
 \end{aligned} \tag{1}$$

Therefore, with probability at least $1 - \delta$, for a given $f \in \mathcal{F}$,

$$|R_S(f) - R(f)| \leq (b-a) \sqrt{\frac{\log \frac{2}{\delta}}{2n}}.$$

The result obtained so far is for one given element $f \in \mathcal{F}$. In order to generalize this result to all functions in $\mathcal{F}$, we rely on the union bound which states that for a family of N functions, the probability of the union of a set of events C_i is upper bounded as:

$$P(C_1 \cup \dots \cup C_N) \leq \sum_{n=1}^N P(C_n) \leq N\delta. \tag{2}$$

Using the union bound for all set of function classes $\mathcal{F} = \{f_1, \dots, f_N\}$, we get that with probability at least $1 - \delta$,

$$\forall f \in \mathcal{F}, R(f) \leq R_S(f) + (b-a) \sqrt{\frac{\log N + \log \frac{2}{\delta}}{2n}}.$$

Indeed, the derivation is similar to what we did above:

$$\begin{aligned}
 \delta &= 2N \exp \left(-\frac{2n\epsilon^2}{(b-a)^2} \right) \\
 \implies \log \delta &= \log 2N - \left(\frac{2n\epsilon^2}{(b-a)^2} \right) \\
 \implies \log \frac{1}{\delta} &= -\log 2N + \left(\frac{2n\epsilon^2}{(b-a)^2} \right) \\
 \implies \epsilon^2 &= \frac{(b-a)^2}{2n} \left(\log 2N + \log \frac{1}{\delta} \right) \\
 \implies \epsilon &= (b-a) \sqrt{\frac{\log \frac{2N}{\delta}}{2n}}
 \end{aligned} \tag{3}$$

Remark 1 (Uncountable case). *When the set $\mathcal{F}$ is uncountable, the previous approach does not directly work. There are several measures of capacity or size of classes of function: the VC dimension and growth function both distribution independent, and the VC entropy which depends on the distribution.*

2 Uniform Bounds

Although the above result seems nice (since it applies to any class of bounded functions), it is actually severely limited. Indeed, what it essentially says is that for each (fixed) function $f \in \mathcal{F}$, there is a collection of sample sets S for which the bound holds, and this collection of sample sets has measure $\mathbb{P}[S] \geq 1 - \delta$. However, these sets S may be different for different functions f . In other words, for the observed sample, only some of the functions in $\mathcal{F}$ will satisfy the bound. This means that f cannot change with different draws of S for the bound to hold.

Note that for simplicity, we will consider the case where $(b-a) = 1$.

Before seeing the data, we do not know which function f the algorithm will choose. The idea is to consider uniform bounds, i.e. bounds that hold uniformly over all functions in the class, hence also for the supremum (i.e. the worst-case),

$$\sup_{f \in \mathcal{F}} R(f) - R_S(f). \quad (4)$$

Remark 2 (Why the first proof breaks down). *The argument in the previous section implicitly assumes that the function f is chosen independently of the data sample S . In other words, the bound is derived for a fixed f decided before observing any data.*

Formally, Hoeffding's inequality controls the deviation of a fixed random variable—here, the empirical average of $f(Z_i)$ —from its expectation. However, in machine learning, the algorithm selects f_S after seeing the sample S . Hence, f_S depends on the data and is itself a random function.

Once f depends on S , the quantities $f(Z_i)$ are no longer independent and identically distributed, violating the assumptions required by Hoeffding's inequality. Therefore, the first proof cannot be applied to data-dependent choices of f , motivating the need for uniform bounds that hold simultaneously for all functions in $\mathcal{F}$.

Theorem 1. *Given a set S of size n and $\mathcal{F} = \{f_1, \dots, f_N\}$, let $f^* = \operatorname{argmin}_{f \in \mathcal{F}} R(f)$ and $f_S = \operatorname{argmin}_{f \in \mathcal{F}} R_S(f)$. Then*

$$R(f_S) \leq R(f^*) + 2\sqrt{\frac{\log N + \log \frac{2}{\delta}}{2n}}. \quad (5)$$

Proof. For each function f_i , we define a set of n -tuples:

$$C_i = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n) \mid R(f_i) - R_S(f_i) > \epsilon\} \quad i = 1, 2, \dots, N. \quad (6)$$

Note that the set C_i contains all the "bad" samples, i.e. those for which the bound fails.

For each function, we can use Hoeffding's inequality to predict that

$$P(C_i) \leq \delta \quad (7)$$

and therefore

$$P(C_1 \cup C_2) \leq P(C_1) + P(C_2) \leq 2\delta \quad (8)$$

which can be generalized to a family of N functions:

$$P(C_1 \cup \dots \cup C_N) \leq \sum_{n=1}^N P(C_n) \leq N\delta. \quad (9)$$

We thus have that for any family of N functions,

$$P(\sup_{f \in \mathcal{F}} R(f) - R_S(f) \geq \epsilon) \leq Ne^{-2n\epsilon^2}, \quad (10)$$

or equivalently that if $\mathcal{F} = \{f_1, \dots, f_N\}$, with probability $1 - \delta$,

$$\forall f \in \mathcal{F}, R(f) \leq R_S(f) + \sqrt{\frac{\log N + \log \frac{2}{\delta}}{2n}}. \quad (11)$$

Now, we need to derive a bound for the minimum of the empirical risk f_S , i.e. $R_S(f_S) = \min_{f \in \mathcal{F}} R_S(f)$. This is done as follows.

Recall that $f^* = \operatorname{argmin}_{f \in \mathcal{F}} R(f)$. Starting from the inequality

$$R(f^*) = R(f^*) - R_S(f^*) + R_S(f^*) \leq R_S(f^*) + \sup_{f \in \mathcal{F}} (R(f) - R_S(f)). \quad (12)$$

Combined with $R_S(f^*) - R_S(f_S) \geq 0$ by definition of f_S , we get

$$\begin{aligned} R(f_S) &= R(f_S) \pm R(f^*) \leq \underbrace{R_S(f^*) - R_S(f_S)}_{\geq 0} + R(f_S) - R(f^*) + R(f^*) \\ &\leq 2 \sup_{f \in \mathcal{F}} |R(f) - R_S(f)| + R(f^*). \end{aligned} \quad (13)$$

Hence with probability at least $1 - \delta$,

$$R(f_S) \leq R(f^*) + 2\sqrt{\frac{\log N + \log \frac{2}{\delta}}{2n}}. \quad (14)$$

□

3 Infinite Case: VC Bound

The previous result can also be extended to the case where the class $\mathcal{F}$ is infinite using VC theory. Vapnik & Chervonenkis argued that what really matters for a sample $S = \{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ containing n points is the set

$$\mathcal{F}_{\mathbf{z}_1, \dots, \mathbf{z}_n} = \{(f(\mathbf{z}_1), \dots, f(\mathbf{z}_n)) | f \in \mathcal{F}\}. \quad (15)$$

The size of this set is the total number of possible ways that S can be classified. For binary classification, $\mathcal{F}_{\mathbf{z}_1, \dots, \mathbf{z}_n}$ is a set of binary vectors $\subset \{-1, +1\}^n$, see Figure 1 below.

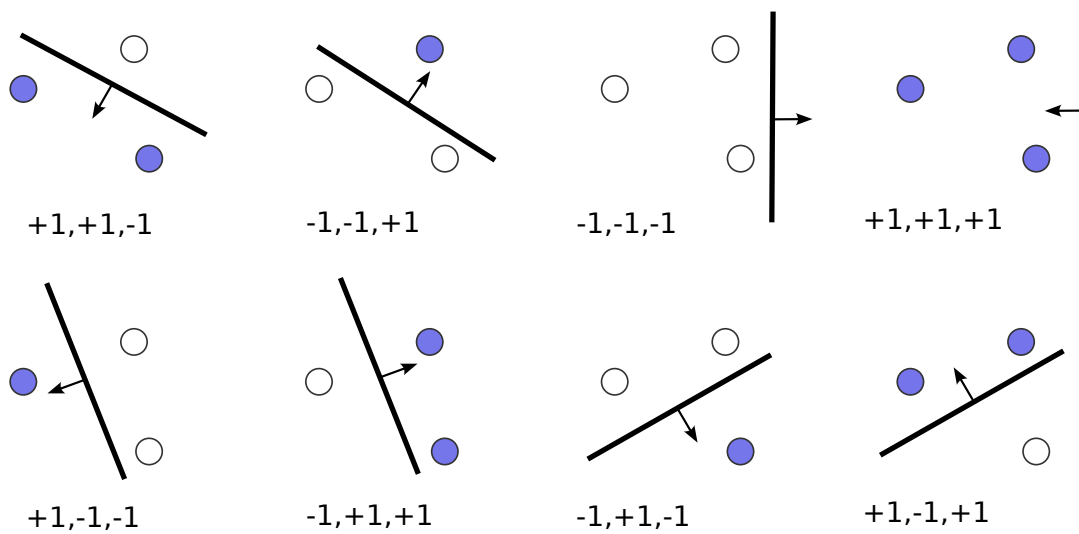


Figure 1: Illustration of the concept of growth function.

Definition 1 (Growth Function). *The growth function of $\mathcal{F}$ is equal to*

$$S_{\mathcal{F}}(n) = \sup_{(\mathbf{z}_1, \dots, \mathbf{z}_n)} |\mathcal{F}_{\mathbf{z}_1, \dots, \mathbf{z}_n}|. \quad (16)$$

The growth function can be thought as a measure of the "size" for the class of functions $\mathcal{F}$. Indeed, the size of this set is the number of possible ways in which the data $\{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ can be classified using functions in $\mathcal{F}$. An upper bound on this number is 2^n .

Example Consider the set of halfspaces in 2-dimensions, i.e binary functions whose decision boundary is a line. One can easily see that the growth function for one, two, and three points are $2^1, 2^2, 2^3$. However, this exponential trend does not continue past 4 points (exercise: check this yourself).

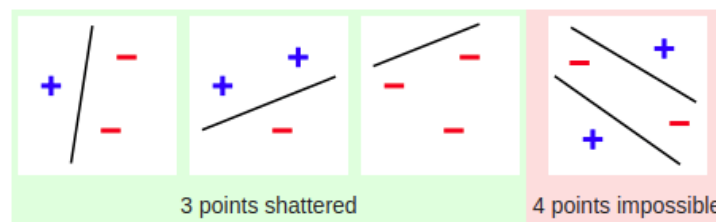


Figure 2: Illustration of the concept of shattering. Source: Wikipedia.

We are now ready to state the VC theorem that gives us a generalization bound that depends on the growth function.

Theorem 2. For any $\delta > 0$, with probability at least $1 - \delta$,

$$\forall f \in \mathcal{F}, \quad R(f) \leq R_S(f) + 2\sqrt{2 \frac{\log S_{\mathcal{F}}(2n) + \log \frac{2}{\delta}}{n}},$$

where $S_{\mathcal{F}}(2n)$ is the growth function of the function class $\mathcal{F}$ defined as

$$S_{\mathcal{F}}(n) = \sup_{\mathbf{z}} |\mathcal{F}_{\mathbf{z}_1, \dots, \mathbf{z}_n}|.$$

Proof. Proof omitted. See statistical theory book. □

VC dimension An important concept in VC theory is shattering. We say that $\mathcal{F}$ shatters S if $|\mathcal{F}_S| = 2^{|S|}$. The VC dimension of a function class $\mathcal{F}$ is the cardinality of the largest set that it can shatter. Importantly this means that we can place the points in the set S arbitrarily, we only need one configuration of these points that can be shattered. Formally, we will define the VC dimension as follows.

Definition 2 (VC dimension). The VC dimension of a class $\mathcal{F}$ is the largest n such that

$$S_{\mathcal{F}}(n) = 2^n.$$

Since n points can have 2^n configurations, the VC dimension is the largest number of points that can be shattered (i.e. split arbitrarily) by the function class. For example, the VC dimension of linear classifiers in $\mathbb{R}^d$ is $d + 1$ (see Figure 2). How do we prove this? We need to show there exists a configuration of $d + 1$ points that can be shattered, but that there does not exist a configuration of $d + 2$ points that can be shattered.

The following well-known lemma establishes an upper bound on the growth function.

Lemma 3 (Sauer-Shelah lemma). Let $\mathcal{F}$ be a class of functions with finite VC-dimension h . Then, $\forall n \in \mathbb{N}$,

$$S_{\mathcal{F}}(n) \leq \sum_{i=0}^h \binom{n}{i}, \tag{17}$$

and for all $n \geq h$,

$$S_{\mathcal{F}}(n) \leq \left(\frac{en}{h}\right)^h. \tag{18}$$

The first bound shows that the growth function grows very fast as a function of n (faster than exponential), but once the number of datapoints n is larger than the VC dimension h , the growth becomes polynomial.

One can also use this upper bound in combination with Theorem 2 in order to obtain the following generalization bound that depends on h and n :

$$\forall f \in \mathcal{F}, \quad R(f) \leq R_S(f) + 2\sqrt{2 \frac{h \log \frac{2en}{h} + \log \frac{2}{\delta}}{n}}.$$

Application to neural networks Bartlett et al. (2019) proves bounds on the VC dimension of piecewise linear networks. Letting W be the number of weights and L be the number of layers, they prove that the VC dimension is $\mathcal{O}(WL \log(W))$. VC bounds therefore claim that the number of datapoints required to obtain a small generalization gap has to scale inversely proportionally to $\mathcal{O}(WL \log(W))$. From a practical point of view, these bounds tend to be too large for modern deep neural networks that rely on a large number of parameters. This is partly due to their generality as VC bounds provide guarantees for any probability distribution and for any training algorithm. We will later discuss other alternatives that can yield practically more useful generalization bounds but it is worth emphasizing at this stage that the generalization of deep neural networks is an active area of research.

4 Rademacher Bound

A Rademacher random variable σ_i is defined as

$$\sigma_i = \begin{cases} +1, & \text{with prob. } 1/2 \\ -1, & \text{with prob. } 1/2 \end{cases}$$

Definition 3 (Rademacher complexity). Let $S = \{z_1, \dots, z_n\}$ be a set of samples drawn i.i.d. from D . Let $\mathcal{F}$ be a class of functions $f : Z \rightarrow \mathbb{R}$. The empirical Rademacher complexity is then defined as

$$\hat{\mathbb{R}}_S(\mathcal{F}) = \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(z_i) \right]. \quad (19)$$

The supremum measures the maximum correlation between $f(z_i)$ and σ_i over all $f \in \mathcal{F}$. Taking the expectation over σ , we can say that the empirical Rademacher complexity measures the ability of functions from $\mathcal{F}$ to fit random noise.

The expected Rademacher complexity is then obtained by taking the expectation of $\hat{\mathbb{R}}_S(\mathcal{F})$ over all samples of size n , i.e.

$$\mathbb{R}_n(\mathcal{F}) = \mathbb{E}_S[\hat{\mathbb{R}}_S(\mathcal{F})], \quad (20)$$

which measures the expected noise fitting ability of $\mathcal{F}$ over all datasets S .

In the following, we will use the notation $\hat{\mathbb{E}}_S[f] = \frac{1}{n} \sum_{i=1}^n f(z_i)$ where $S = \{z_1, \dots, z_n\}$. The following result states a generalization bound in terms of the Rademacher complexity of a given function class.

Theorem 4 (Rademacher-based uniform convergence). Fix the distribution D and $\delta \in (0, 1)$. If $\mathcal{F} \in \{f : Z \rightarrow [0, 1]\}$ and $S = \{z_1, \dots, z_n\}$. Then with probability at least $1 - \delta$ over the draw of S , for all $f \in \mathcal{F}$, we have

$$\mathbb{E}_D[f(z)] \leq \hat{\mathbb{E}}_S[f(z)] + 2\mathbb{R}_n(\mathcal{F}) + \sqrt{\frac{\log(1/\delta)}{2n}}. \quad (21)$$

Proof sketch. We only state the main steps:

1. Define $\phi(S) = \sup_{f \in \mathcal{F}} (\mathbb{E}_D[f] - \hat{\mathbb{E}}_S[f])$. Then

$$\mathbb{E}_D[f(z)] \leq \hat{\mathbb{E}}_S[f(z)] + \phi(S)$$

2. Bound $\phi(S)$ by its expected value $\mathbb{E}_S[\phi(S)]$ using McDiarmid's inequality². McDiarmid's Inequality provides a concentration bound on the deviation of the sample mean of independent random variables from its expected value.
3. Define a ghost sample $S' = \{z'_1, \dots, z'_n\} \sim D$ and show $\mathbb{E}_S[\phi(S)] \leq \mathbb{E}_{S, S'}[\sup_{f \in \mathcal{F}} (\hat{\mathbb{E}}_{S'}[f] - \hat{\mathbb{E}}_S[f])]$
 $\rightsquigarrow$ To prove this, note that $\mathbb{E}_D[f] = \mathbb{E}_{S'}[\hat{\mathbb{E}}_{S'}[f]]$ and $\sup_{f \in \mathcal{F}} \mathbb{E}[f] \leq \mathbb{E}[\sup_{f \in \mathcal{F}} f]$
4. Show $\mathbb{E}_{S, S'}[\sup_{f \in \mathcal{F}} (\hat{\mathbb{E}}_{S'}[f] - \hat{\mathbb{E}}_S[f])] = \mathbb{E}_{S, S'}[\sup_{f \in \mathcal{F}} \sum_i \sigma_i (f(z'_i) - f(z_i))]$
 $\rightsquigarrow$ To prove this, use the fact that z'_i and z_i are interchangeable
5. Show $\mathbb{E}_{S, S'}[\sup_{f \in \mathcal{F}} \sum_i \sigma_i (f(z'_i) - f(z_i))] \leq 2\mathbb{R}_n(\mathcal{F})$
 $\rightsquigarrow$ To prove this, break the sum into two sums, one over z_i and the other over z'_i

□

Finally, we note that we can translate the result from the function class $\mathcal{F}$ to the final loss using the following lemma, whose proof is left as an exercise.

Lemma 5. Let $\mathcal{F}$ be a family of functions taking values in $\{-1, +1\}$ and let $\mathcal{G}$ be the family of loss functions associated to $\mathcal{F}$ for the zero-one loss: $\mathcal{G} = \{(x, y) \mapsto \mathbf{1}_{f(x) \neq y} : f \in \mathcal{F}\}$. For any sample $S = ((x_1, y_1), \dots, (x_n, y_n))$ of elements in $\mathcal{X} \times \{-1, +1\}$, let $S_{\mathcal{X}}$ denote its projection over $\mathcal{X}$: $S_{\mathcal{X}} = (x_1, \dots, x_n)$. Then, the following relation holds between the empirical Rademacher complexities of $\mathcal{G}$ and $\mathcal{F}$:

$$\mathbb{R}_S(\mathcal{G}) = \frac{1}{2} \mathbb{R}_{S_{\mathcal{X}}}(\mathcal{F}). \quad (22)$$

Proof. Exercise. □

²McDiarmid's Inequality: Let $X_1, X_2, \dots, X_n$ be independent random variables, and let $f : \mathcal{X}^n \rightarrow \mathbb{R}$ be a function. If there exist constants $c_1, c_2, \dots, c_n > 0$ such that for all $i = 1, 2, \dots, n$ and $x_i, x'_i \in \mathcal{X}$: $|f(x_1, x_2, \dots, x_i, \dots, x_n) - f(x_1, x_2, \dots, x'_i, \dots, x_n)| \leq c_i$, then for any $\epsilon > 0$:

$$\Pr \left(\left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X)] \right| \geq \epsilon \right) \leq 2e^{-\frac{2\epsilon^2}{\sum_{i=1}^n c_i^2}}.$$

Theorem 6 (Rademacher complexity bounds – binary classification). *Let $\mathcal{F}$ be a family of functions taking values in $\{-1, +1\}$ and let $\mathcal{D}$ be the distribution over the input space $\mathcal{X}$. Then, for any $\delta > 0$, with probability at least $1 - \delta$ over a sample S of size n drawn according to $\mathcal{D}$, each of the following holds for any $h \in \mathcal{H}$:*

$$R(h) \leq \hat{R}_S(h) + \mathbb{R}_n(\mathcal{F}) + \sqrt{\frac{\log \frac{1}{\delta}}{2n}}. \quad (23)$$

Proof. Combine Theorem 4 and Lemma 5. □

Rademacher bound for Linear Classes

Lemma 7. *Let $\mathcal{F}$ be the class of linear predictors, with the L_2 -norm of the weights bounded by a constant $W_2 > 0$. Also assume that with probability one that $\|\mathbf{x}\|_2 \leq D$. Then*

$$\mathbb{R}(\mathcal{F}) \leq \frac{DW_2}{\sqrt{n}}. \quad (24)$$

Proof. Let $\mathcal{F}$ be the class

$$\mathcal{F} = \{(\mathbf{w}^\top \mathbf{x}_1, \mathbf{w}^\top \mathbf{x}_2, \dots, \mathbf{w}^\top \mathbf{x}_n) : \|\mathbf{w}\|_2 \leq W_2\}.$$

The empirical Rademacher complexity of $\mathcal{F}$ can be bounded as follows:

$$\begin{aligned} \mathbb{R}(\mathcal{F}) &= \mathbb{E}_\sigma \left[\frac{1}{n} \sum_{i=1}^n \sup_{f \in \mathcal{F}} \sigma_i f(z_i) \right] \\ &= \frac{1}{n} \mathbb{E}_\sigma \left[\sup_{\mathbf{w}: \|\mathbf{w}\|_2 \leq W_2} \sum_{i=1}^n \sigma_i \mathbf{w}^\top \mathbf{x}_i \right] \\ &= \frac{1}{n} \mathbb{E}_\sigma \left[\sup_{\mathbf{w}: \|\mathbf{w}\|_2 \leq W_2} \mathbf{w}^\top \left(\sum_{i=1}^n \sigma_i \mathbf{x}_i \right) \right]. \end{aligned} \quad (25)$$

Since $\|\mathbf{w}\|_2 \leq W_2$ for all $\mathbf{w}$, then

$$\begin{aligned} \mathbb{R}(\mathcal{F}) &\leq \frac{W_2}{n} \mathbb{E}_\sigma \left[\left\| \sum_{i=1}^n \sigma_i \mathbf{x}_i \right\|_2 \right] \\ &\leq \frac{W_2}{n} \sqrt{\mathbb{E}_\sigma \left[\sum_{i=1}^n \|\sigma_i \mathbf{x}_i\|_2^2 \right]} \\ &= \frac{W_2}{n} \sqrt{\mathbb{E}_\sigma \left[\sum_{i=1}^n \sigma_i^2 \|\mathbf{x}_i\|_2^2 \right]} \\ &= \frac{W_2}{n} \sqrt{\mathbb{E}_\sigma \left[\sum_{i=1}^n \|\mathbf{x}_i\|_2^2 \right]} \\ &\leq \frac{DW_2}{\sqrt{n}}. \end{aligned} \quad (26)$$

□

Note that in the bound derived in Lemma 7, the Lipschitz constant of f w.r.t. to $\mathbf{x}$ appears, since $\|\nabla_{\mathbf{x}} f(\mathbf{z}_i)\|_2 = \|\mathbf{w}\|_2 \leq W_2$.

Another important lemma that is often used to bound the Rademacher complexity of composition of functions is the composition lemma stated below.

Lemma 8 (Composition lemma). *For $\mathcal{F} \in \mathbb{R}^d$, $\phi : \mathbb{R} \rightarrow \mathbb{R}$, let $\mathcal{F}' := \{\phi \circ f : f \in \mathcal{F}\}$. If ϕ is L -Lipschitz continuous, i.e. $|\phi(t) - \phi(t')| \leq L|t - t'|$, then for any m ,*

$$\mathbb{R}(\mathcal{F}') \leq L\mathbb{R}(\mathcal{F}). \quad (27)$$

Proof. Exercise. □

Application to neural networks Based on Lemma 7 and 8, we can derive a bound on the Rademacher complexity of a one-layer neural network with a Lipschitz activation function, i.e. $f(\mathbf{x}) = \sigma(\mathbf{W}\mathbf{x})$ where σ is Lipschitz. These results can also be extended to multiple layers by relating the Rademacher complexity of a layer to its previous layer (using a Contraction argument) and then use a bound similar to the one above for the first layer, see e.g. details in Neyshabur et al. (2015). This leads to the following theorem.

Theorem 9. Consider an L -layer ReLU network with activation function $\sigma(t) = \max(0, t)$ with bounded norm:

$$\mathcal{F}_B := \{\mathbf{x} \rightarrow \mathbf{W}_L \sigma(\mathbf{W}_{L-1} \dots \sigma(\mathbf{W}_1 \mathbf{x}) \dots), \mathbf{W}_k \in \mathbb{R}^{m_k \times m_{k-1}}, m_0 = \dim(\mathbf{x}), m_L = 1, m_k \in \mathbb{N}, \|\mathbf{W}_i\|_F \leq B \forall i\}.$$

Then for a training set $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n) \in \mathcal{X} \times \{\pm 1\}$ with $\|\mathbf{x}_i\| \leq 1$, we have that a.s.

$$\mathbb{R}(\mathcal{F}_B) \leq \frac{(2B)^L}{\sqrt{n}}.$$

As a word of caution, existing Rademacher bounds typically depend on the product of the weight matrices of a neural network, which lead to loose or even vacuous bounds (i.e. they have no predictive abilities). Compression arguments such as in Arora et al. (2018) can make these bounds tighter but this is very much an active area of research where significant advances are required to obtain practically useful bounds. To make things worse, the work of Zhang et al. (2021) has shown empirically that for certain datasets, neural networks can fit random labels with zero training error. This indicates that these networks can maximize the Rademacher complexity, therefore highlighting some potentially severe shortcomings of such approaches. In the next lecture, we will see an alternative approach based on PAC-Bayes bounds that also have long-standing theoretical roots but are also currently seen as empirically promising to evaluate the generalization ability of deep neural networks.

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